AP PHYSICS 1

Unit 1 • Worksheet 1.3A

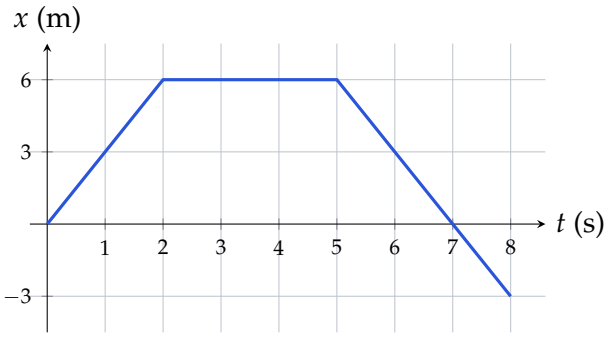
Worksheet 1.3A • Representing Motion

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Questions 1 to 4 refer to the graph below: the position-time graph of an object moving along a line.



Question 1

What is the velocity of the object between $t = 0$ and $t = 2\,\text{s}$?

(A) $6\,\text{m/s}$
(B) $3\,\text{m/s}$
(C) $2\,\text{m/s}$
(D) 0

Question 2

Between $t = 2\,\text{s}$ and $t = 5\,\text{s}$, the object is

(A) at rest
(B) moving at a constant positive velocity
(C) speeding up
(D) moving backward

Question 3

What is the velocity of the object between $t = 5\,\text{s}$ and $t = 8\,\text{s}$?

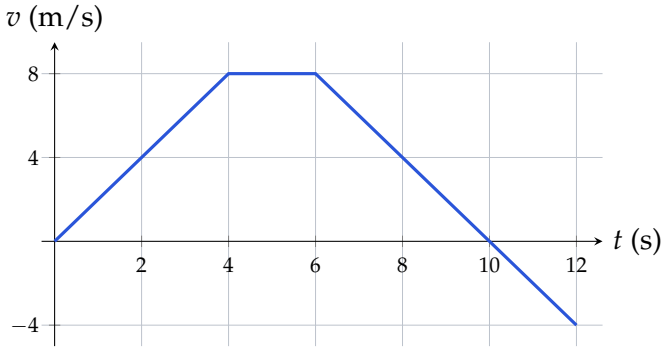
(A) $3\,\text{m/s}$
(B) $-9\,\text{m/s}$
(C) $-3\,\text{m/s}$
(D) $1\,\text{m/s}$

Question 4

What is the displacement of the object from $t = 0$ to $t = 8\,\text{s}$?

(A) $6\,\text{m}$
(B) $-3\,\text{m}$
(C) $15\,\text{m}$
(D) $3\,\text{m}$

Questions 5 to 8 refer to the graph below: the velocity-time graph of a cart moving along a line.



Question 5

What is the acceleration of the cart at $t = 2\,\text{s}$?

(A) $8\,\text{m/s}^2$
(B) $4\,\text{m/s}^2$
(C) $2\,\text{m/s}^2$
(D) 0

Question 6

What is the displacement of the cart from $t = 0$ to $t = 6\,\text{s}$?

(A) $32\,\text{m}$
(B) $48\,\text{m}$
(C) $16\,\text{m}$
(D) $24\,\text{m}$

Question 7

At what time does the cart change direction?

(A) 6 s
(B) 4 s
(C) 12 s
(D) 10 s

Question 8

What is the displacement of the cart from $t = 0$ to $t = 12\,\text{s}$?

(A) 52 m
(B) 48 m
(C) 44 m
(D) 40 m

Question 9 (Free response)

A cart starts from rest at $x = 0$. It accelerates at $2\,\text{m/s}^2$ for 3 s, then moves at constant velocity for 3 s.

(a) On the first axes, sketch the velocity-time graph from $t = 0$ to $t = 6\,\text{s}$. Label the values.
(b) On the second axes, sketch the position-time graph. Describe the shape of each part.
(c) Find the displacement at $t = 6\,\text{s}$.

